PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Economic evaluations of musculoskeletal physiotherapy: Protocol of a systematic review
AUTHORS	Baumbach, Linda; König, Hans-Helmut; Kretzler, Benedikt; Hajek, André

VERSION 1 – REVIEW

REVIEWER	Jowett, Sue University of Birmingham, Health Economics Unit
REVIEW RETURNED	05-Nov-2021

GENERAL COMMENTS	Overall this is a clear protocol of a proposed review. I have 4 minor comments for the authors to consider 1) In the methods it states papers will be considered up to 2021. It would be good to clarify which month. 2) Will conference abstracts be excluded? If so this needs to be included in the exclusion criteria 3) Do you plan to present more detail on the models you find e.g. type of model? 4) Will you present your results tables/finding by musculoskeletal condition? This would be useful for readers, for example to look at the results for back pain together.

REVIEWER	Kim, Nam-kwen Pusan National University Korean Medicine Hospital
REVIEW RETURNED	15-Nov-2021

GENERAL COMMENTS	This manuscript is about systematic review of economic evaluation studies for musculoskeletal physiotherapies. This systematic evaluation research hired the reasonable methodologies and described properly by following the reporting guideline of PRISMA-P. I recommend this manuscript could be published after minor revisions (correction and consideration), which are as follows. Minor Revision 1. In line 106: trough – through? 2. In Table 1 Intervention: Physiotherapies only group and additional physiotherapies should be addressed in the Intervention criteria. Combination treatments (usual care or else) could affect the results and conclusions, and so, the author should address this issue by using subgroup analyses or else. I recommend above minor revision for improving the author's research results.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Sue Jowett, University of Birmingham

Reviewer comment
Overall this is a clear protocol of a proposed review. I have 4 minor comments for the authors to consider
Author response
We like to thank the reviewer for the comments which helped us to enhance our Review protocol.

Reviewer comment
1) In the methods it states papers will be considered up to 2021. It would be good to clarify which month.
Author response
We like to thank you for your criticism and clarified in our manuscript that we searched up to the 31st of January 2022.
Author action
Page 4 line 97, 98
We will include full economic evaluation studies of musculoskeletal physiotherapy, published in any language up to the 31st of January 2022.

Reviewer comment
2) Will conference abstracts be excluded? If so this needs to be included in the exclusion criteria
Author response
We like to thank you for your valid input, indeed we do not include conference abstracts and added this to the exclusion criteria.
Author action
Page 5 Table 1. Under the exclusion criteria regarding the type of studies
“Conference abstracts and studies without results eg. Protocols, Review studies”

Reviewer comment
3) Do you plan to present more detail on the models you find e.g. type of model?
Author response
We appreciate your comment, helping us to improve and clarify our manuscript. We will provide an overview of the different types of analyses, which have been used in the included studies.
Author action
Page 9 Line 179 - 184
The results of the included studies will be synthesized by different characteristics. We will provide an overview of the publication years, countries of origin, type of analyses (clinical trial vs. modelling with a stratification for decision tree, Markov model, discrete event simulation), time of interventions (primary, secondary, or tertiary prevention), over all types of interventions (primary or add-on, eg. after surgery), and the specific musculoskeletal condition. A special focus will be set on the latter one, i.e., the summary of results of the included studies including a stratification for specific type of intervention will be provided by musculoskeletal condition.

Reviewer comment
4) Will you present your results tables/finding by musculoskeletal condition? This would be useful for readers, for example to look at the results for back pain together
Author response
We like to thank you, for highlighting this uncertainty. We will synthesize the studies based on the different characteristics (publication years, countries of origin, type of analyses, and the specific

musculoskeletal condition). We agree that the synthesis on specific musculoskeletal condition is most interesting to the readers. Thus, we will pay special attention to this and we will provide tables accordingly.
Author action
Page 9 Line 179 - 184
The results of the included studies will be synthesized by different characteristics. We will provide an overview of the publication years, countries of origin, type of analyses (clinical trial vs. modelling with a stratification for decision tree, Markov model, discrete event simulation), time of interventions (primary, secondary, or tertiary prevention), over all types of interventions (primary or add-on, eg. after surgery), and the specific musculoskeletal condition. A special focus will be set on the latter one, i.e., the summary of results of the included studies including a stratification for specific type of intervention will be provided by musculoskeletal condition.

Reviewer: 2
Dr. Nam-kwen Kim, Pusan National University Korean Medicine Hospital

Reviewer comment
This manuscript is about systematic review of economic evaluation studies for musculoskeletal physiotherapies. This systematic evaluation research hired the reasonable methodologies and described properly by following the reporting guideline of PRISMA-P. I recommend this manuscript could be published after minor revisions (correction and consideration), which are as follows.
Author response
Thank you for your helpful comments. It helps to improve the quality of the manuscript.

Reviewer comment
1. In line 106: trough – through?
Author response
Thank you for noting this typo. We corrected it accordingly:
Author action
Page 5 Line 107
• Medline (through PubMed)

Reviewer comment
2. In Table 1 Intervention: Physiotherapies only group and additional physiotherapies should be addressed in the Intervention criteria. Combination treatments (usual care or else) could affect the results and conclusions, and so, the author should address this issue by using subgroup analyses or else.
Author response
Thank you for your comment. We agree that it remains unclear that we indeed only include studies with interventions, which can be conducted by physiotherapists alone. Please note that physiotherapy is a profession which offers different treatments. Thus, the intervention of a physiotherapist is likely to consist of different treatment modalities. In this review, we are interested in economic evaluations comparing different treatment modalities within the physiotherapeutic intervention and also in economic evaluations comparing physiotherapeutic intervention with another intervention such as usual care or surgery. Thus, economic evaluations of multidisciplinary interventions or combination treatments will be excluded. However, studies in which the control and the intervention are offered or receive an additional treatment such as pain medication or surgery will still be included. We clarified this in Table 1.
Further, we add to our results that we will inform on the distribution of studies evaluating physiotherapy as “primary, secondary or tertiary prevention” as well as if physiotherapy was provided as add-on (eg. after surgery) or as initial treatment.

Author action
Page 5 Table 1 Under the criteria regarding the type of intervention
Inclusion criteria
Any intervention, which can be conducted by a physiotherapist alone, including amongst other patient education, single or group-based exercise therapy, and manual therapy, any additional treatment would need to be offered/delivered in the intervention and in the control group (eg. pain medication or surgery)
Exclusion criteria
Interventions offered from non-health professionals including sport instructors and
E-interventions or by an interdisciplinary team, including studies in which the physiotherapeutic treatment of interest is combined with treatments mostly delivered from other disciplines.

Page 9 Line 179 - 184
The results of the included studies will be synthesized by different characteristics. We will provide an overview of the publication years, countries of origin, type of analyses (clinical trial vs. modelling with a stratification for decision tree, Markov model, discrete event simulation), time of interventions (primary, secondary, or tertiary prevention), over all types of interventions (primary or add-on, eg. after surgery), and the specific musculoskeletal condition. A special focus will be set on the latter one, i.e., the summary of results of the included studies including a stratification for specific type of intervention will be provided by musculoskeletal condition.

One more time we like to thank you the editor and reviewers for their constrictive feedback which helped us to improve the quality of our systematic review protocol and will enhance the quality of the related review.